Ubaid **Ullah**

Electrical Engineer · AI Engineer

Gyeongsan-si, Gyeongsangbuk-do, Republic of Korea

(+82) 10-5929-1995 | ubaid95@yu.ac.kr | uu95.github.io | uu95 | ubaid-ullah-75504715b

"True progress comes not from fearlessness, but from refusing to stop."

Summary

MULTIMODAL AI Researcher with 5+ years of experience in Artificial Intelligence, Computer Vision, and Cross-Modal Content Generation. Engineering has always been in my blood—whether building electronic systems, developing software, or designing advanced AI models, I thrive on turning ideas into reality. I have a strong passion for learning through teaching, and I'm deeply motivated to explore new concepts, tools, and methods to push the boundaries of research and innovation.

Work Experience

Intelligent Computer Vision Software Lab (ICVSLab), Yeungnam University — *Gyeongsan, S. Korea*

Graduate Researcher — *Mar. 2021 – August, 2026*

- Advanced cross-modal AI research across computer vision, audio, and LLMs.
- Improved AI efficiency and performance by developing models for image generation, video synthesis, and ECG diagnosis, including models up to 66× smaller.
- Built training-free and resource-efficient multimodal pipelines for music-to-image and music-to-video generation.
- Improved generation quality through audio analysis modules for emotion, structure, and semantic tagging.
- Built and curated multimodal datasets, including around 500K paired audio-visual samples.
- Accelerated research evaluation and deployment by developing software tools, demos, and web platforms.
- Showcased creative AI systems through large-scale live demonstrations of music-to-video generation.
- Authored and co-authored papers targeting ICASSP, Interspeech, ACL, IEEE JBHI, and IEEE Access.
- Contributed to NRF Korea research proposal and project application development.
- Supported undergraduate teaching through lab instruction, examinations, and student mentoring.

Institute of Space Technology (IST) — *Islamabad, Pakistan*

Research Assistant — *May 2019 – Nov. 2019*

- Developed a high-precision parts inspection system with up to 0.1 mm error through hardware-to-software integration using Basic computer vision algorithms.
- Achieved efficient CPU-friendly processing suitable for edge-device deployment.
- Improved inspection reliability by validating the system across diverse parts and test conditions.
- Produced SCIE-indexed research output from the project findings.

Rapidev IoT Solutions — *Islamabad, Pakistan*

Intern — *Sep. 2018 – Nov. 2018*

- Improved user onboarding and product accessibility by creating tutorial videos for IoT educational kits.
- Strengthened client support and product adoption by maintaining client relationships and assisting with technical troubleshooting and implementation.

503 Aviation Base Workshop EME — *Rawalpindi, Pakistan*

Intern — *Jul. 2016 – Sep. 2016*

- Supported efficiency and safety improvement planning by developing small-scale engineering ideas for aviation operations.
- Built practical aviation engineering insight through early-stage design work and hands-on exposure to workshop operations and technical processes.

Education

Yeungnam University — *Gyeongsan, Republic of Korea*

Graduate Studies in Electronic Engineering, **GPA: 4.38/4.50** — *Mar. 2021 – Aug. 2026*

- Completed graduate-level coursework in Computer Vision, Deep Learning, and Signal Processing.
- Received advanced academic training in artificial intelligence and electronic engineering.

Foundation University, Islamabad — *Islamabad, Pakistan*

B.Sc. in Electrical Engineering, **GPA: 3.83/4.00** — *Sep. 2014 – Nov. 2018*

- Specialization in Machine Learning, and Signal Processing.
- Completed major coursework in Robotics, DSP, and Control Systems.

Publications

IoT-enabled computer vision-based parts inspection system for SME 4.0 — *Microprocessors and Microsystems, Elsevier*

First Author — *2021*

- **U. Ullah**, F. A. Bhatti, A. R. Maud, M. I. Asim, K. Khurshid, M. Maqsood. Published in *Microprocessors and Microsystems*, vol. 87, 104354.

A review of multi-modal learning from the text-guided visual processing viewpoint — *Sensors (MDPI)*

First Author — *2022*

- **U. Ullah**, J. S. Lee, C. H. An, H. Lee, S. Y. Park, R. H. Baek, H. C. Choi. Published in *Sensors*, vol. 22, no. 18, 6816.

MuIm: Analyzing Music–Image Correlations from an Artistic Perspective — *Applied Sciences (MDPI)*

First Author — *2024*

- **U. Ullah**, H. C. Choi. Published in *Applied Sciences*, vol. 14, no. 23, 11470.

Enhancing Multimodal Facial Expression Recognition with Derived Modality Transformations Using Novel Convolutional Neural Network-Based Models — *IEEE Access*

Co-first Author — *2026*

- M. T. Naseem, **U. Ullah**, C. S. Lee.

Efficient Coarse-to-Fine Learning for Sparse Long-Tailed Multi-Label ECG Classification — *Scientific Reports (under review)*

First Author — *2026*

- **U. Ullah**, H. C. Choi.

Music Artistic Captioning: Towards Translating Music into Expressive Language — *Interspeech Main Track*

First Author — *2026*

- **U. Ullah**, H. C. Choi, Z. Bouraoui.

A brief survey of text-driven image generation and manipulation — *IEEE ICCE-Asia Conference*

Co-Author — *2021*

- H. Lee, **U. Ullah**, J. S. Lee, B. Jeong, H. C. Choi. Published in *IEEE International Conference on Consumer Electronics-Asia*, pp. 1–4.

Real-time Compact Video Super-resolution for Video Conferencing — *The Korean Institute of Electrical Engineers Conference*

First Author — *2022*

- **U. Ubaid**, H. C. Choi. Published in *KIEE Conference Proceedings*, pp. 1021–1024.

Honors & Awards

International

2021–2026 **Fully Funded Graduate Research Scholarship**, Yeungnam University — *South Korea*

Domestic

2018 **Gold Medal**, Bachelor's Degree — *Pakistan*

Extracurricular Activity

Server, Workstation, and Full-PC Setup & Configuration — *South Korea*

Technical Experience

- Hands-on experience in server and full-PC setup, hardware integration, multi-GPU configuration, system installation, and performance-oriented environment management for research and development.
- Worked on practical system-level configuration, deployment, troubleshooting, and optimization for computational and AI workloads.

Automation and IT Development for Premises and Facilities — *Pakistan / South Korea*

Technical Experience

- Experience in automation-oriented IT development for homes and facility environments, including practical setup, integration, and maintenance of technical systems.
- Applied engineering and problem-solving skills to improve usability, reliability, and operational efficiency in real-world environments.

Experience with Top International Research Venues
International

ACADEMIC EXPOSURE

- Gained exposure to the research, review, and presentation standards of leading international venues such as CVPR, ICLR, NeurIPS, and related conferences through active participation and attendance.
- Developed familiarity with high-impact academic communication, publication expectations, and cutting-edge research trends in AI and machine learning.

Domestic and International Interdisciplinary Collaboration
South Korea / France

RESEARCH COLLABORATION

- Collaborated with domestic research groups in South Korea, including interdisciplinary work with art and music departments.
- Engaged in international academic collaboration with researchers in France on research-related projects and publications.